

63. STRENGTH TRAINING DURING COMPETITION PERIOD: YES OR NO?

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In previous KisW Research Notes, we saw that:

- during the preparatory period, 10 weeks of heavy gym strength training improves both short (30s) and long (40min) duration w/kg despite a small increase in body mass (less than 1kg) in elite cyclists. [Read more here](#)
- heavy gym strength training positively impacts also durability (fatigue resistance during very long efforts). [Read more here](#)
- unilateral gym strength training seems better than bilateral to improve cycling performance. [Read more here](#)
- You can read the practical guidelines on how to perform gym strength training to improve cycling performance [here](#).

From a periodization point of view, it is usually easier to perform gym strength training during the preparatory period, given the lower training intensity and lack of competitions. On the other hand, during the competition period, the higher training intensity combined with the presence of competitions and the related travels, make more difficult to perform gym strength training. So, it is common that even many top-level professional cyclists did not perform any gym strength training during the competition period. For example, two case studies analysing training patterns of 5 riders finishing in the top 5 of final general classification of Giro d'Italia and Tour de France between 2015 and 2020 reported that none of the cyclists performed gym strength training during the competition period. (1,2)

Is maintaining gym strength training during the competition period beneficial for road cycling performance?

Let's try to answer using data from two studies from the Norwegian (Lillehammer University) research team lead by Professor Bent Rønnestad.

1. WHAT HAPPENS WHEN GYM STRENGTH TRAINING IS INTERRUPTED DURING THE COMPETITION PERIOD?

The Study:

Rønnestad BR, Hansen J, Hollan I, Spencer M, Ellefsen S. Impairment of Performance Variables After In-Season Strength-Training Cessation in Elite Cyclists. *Int J Sports Physiol Perform.* 2016 Sep;11(6):727-735.

How Did They Train?

- 14 elite cyclists (mean age 20, mean VO2max 75) were divided into a Gym+Bike group (G+B) and a bike only group (B).
- **During the 25 weeks of the preparatory period**, the 2 groups trained similarly (~13.5 hrs/week) on the bike. The only between groups difference was that G+B performed gym heavy strength training sessions (3 × 4-10 RM, 1 to 2 days per week) with the same exercises described [here](#), while B did not perform any gym strength training.

*B performed ~1.5hrs per week more on the bike to match the total G+B training volume.

*RM means “repetitions maximum”. For example, 10RM means doing 10 reps using a weight that permits to do maximum 10 repetitions.

- **When the competition period started**, for 8 weeks, G+B interrupted completely gym strength training and performed similar training to the ‘bike only’ group.

What Did They Find?

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- **Adaptations during the 25 weeks preparatory favoured G+B compared to B only:**
 - 30s maximal mean power (W/Kg): $+2\pm5\%$ vs. $-2\pm4\%$
 - maximal aerobic power output (W/Kg): $+3\pm3\%$ vs. $-3\pm6\%$
 - power output at 4mmol/L (W/kKg): $+4\pm3\%$ vs. $-5\pm14\%$
 - **However, from pre to post the 8 weeks of the competition period (so comparing week 25 vs week 33), the physiological and performance parameters decreased more in G+B compared to B only:**
 - 30s maximal mean power (W/Kg): $-4\pm4\%$ vs. $+1\pm2\%$
 - maximal aerobic power output (W/Kg): $-2\pm4\%$ vs. $+4\pm7\%$
 - power output at 4mmol/L (W/Kg): $0\pm6\%$ vs. $+9\pm14\%$

In addition, some thigh muscle mass gain obtained during the preparatory period only by G+B group, came back to baseline after 8 weeks of stopping gym training.

Doing easy maths, we can appreciate that after all the protocol (week 1 vs week 33) the changes in performance and physiological parameters were very similar between the two groups. **So all the superior cycling related physiological and performance improvement obtained by heavy gym strength training are nullified after only 8 weeks stopping to hit the gym.**

Let's see now the opposite scenario: what happens if strength training is maintained even during the competition period?

2. WHAT HAPPENS WHEN GYM STRENGTH TRAINING IS MAINTAINED EVEN DURING THE COMPETITION PERIOD?

The Study

Rønnestad BR, Hansen EA, Raastad T. In-season strength maintenance training increases well-trained cyclists' performance. Eur J Appl Physiol. 2010 Dec;110(6):1269-82.

How Did They Train?

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- 12 well-trained amateur cyclists (VO_{2max} ~66, age ~30) were divided into a gym+bike group (G+B) and a bike only group (B) that trained similarly for 12 weeks of the **preparatory period**, with the only difference that G+B performed 2xweek gym heavy strength training sessions structured again as those described [here](#).

- When competition period started, for 13 weeks, B continued to train only on the bike, while G+B maintain one gym strength training session but reduced the frequency and load of these sessions: 1x week, 1/2 x 5 @80-85% 1RM.

What Did They Find?

- **Performance adaptations during the 12 weeks preparatory favoured G+B compared to B only:**

Changes in 40min Time Trial Power between week 1 and week 12:

+8±2% vs +4±1%

- **Differently from study 1, in this case the performance advantage obtained during the preparatory period by 'G+B' vs 'B only' were retained and even amplified after 13 weeks of the competition period (25 weeks from the start of the study):**

Changes in 40min Time Trial Power between week 1 and week 25:

+14±3% vs. +4±1%,

In this case, also the thigh muscle mass gains obtained in the preparatory period by the G+B group were retained after 13 weeks of competition period.

STUDY 1+STUDY 2 PRACTICAL TAKEAWAYS:

- **Stopping strength training during the competition period nullifies the cycling performance adaptations previously obtained in the gym during the preparation period.**
- **During the competition period, 1 session per week with low volume per session is enough to maintain the previously obtained gains and further improve performance.**
- The nullified or retained positive performance adaptations seem to be linked to maintaining or losing thigh muscle mass.

GABRIELE'S COMMENT

- **These data globally suggest that one gym strength training session per week even during the competition period should be maintained to optimize performance.**
- **Of course, this should be carefully inserted within the competition period microcycles (which are likely to include a lot of high intensity work both from training sessions and race) to allow the right balance between load and recovery to optimize race performance.**

Thanks for reading! Share with your team!

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